

Maria Tzelepi

Postdoctoral Researcher

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Education

- 03/2021 **PhD**, in *Informatics*, Aristotle University of Thessaloniki (AUTH), Deep Learning Techniques in Digital Media
Supervisor: Prof. Anastasios Tefas
- 09/2016 **MSc**, in *Informatics*, Aristotle University of Thessaloniki (AUTH), Digital Media - Computational Intelligence
- 03/2013 **BSc**, in *Informatics*, Aristotle University of Thessaloniki (AUTH)

Research & Professional Experience

- 2023-Present **PostDoc Researcher**, Information Technologies Institute (ITI) / Centre for Research and Technology Hellas (CERTH), Thessaloniki, Greece
Projects: TRANSMIXR (HORIZON), ENCIRCLE (HORIZON)
- 2021-2023 **PostDoc Researcher**, Aristotle University of Thessaloniki (AUTH), Thessaloniki, Greece
Projects: OpenDR (HORIZON), MALENA (EPAnEK 2014-2020), DeepFinance (EPAnEK 2014-2020), DeepStream (OP PKM 2014-2020)
- 2017-2021 **Researcher**, Aristotle University of Thessaloniki (AUTH), Thessaloniki, Greece
Projects: OpenDR (HORIZON), MULTIDRONE (HORIZON), Lightweight Deep Learning Models for Signal and Information Analysis (OP Human Resources Development, Education and Lifelong Learning 2014-2020), Deep Learning Techniques in Digital Media (GSRT/HFRI)

Awards & Scholarships

- 2022 **Award**, *Greek Women in STEM*, Category: Technology, She STEMs symposium, Greece
- 2022 **Award**, *Best 2020-2021 Doctoral Thesis in the Area of Artificial Intelligence*, Hellenic Artificial Intelligence Society (EETN), Greece
- 2017 **Scholarship**, General Secretariat for Research and Technology (GSRT) and Hellenic Foundation for Research and Innovation (HFRI), 1st Call for HFRI Scholarships to PhD candidates, Greece

Supervisory Experience

- 2019-2023 **Co-supervision**, 6 undergraduate theses & 2 postgraduate master theses
Co-supervised with Prof. Anastasios Tefas (Dept. of Informatics, Aristotle University of Thessaloniki)

Conference Organizing Committee

- 2022 **IEEE Image, Video, and Multidimensional Signal Processing Workshop (IVMSP)**, *Finance Chair*, Nafplio, Greece

Workshop Organization

- 2023 **Open Workshop “AI in Energy”**, in the context of MALENA Project, Co-organization Thessaloniki, Greece

Service

Reviewer, IEEE Transactions on Pattern Analysis and Machine Intelligence, IEEE Transactions on Image Processing, IEEE Transactions on Neural Networks and Learning Systems, Neurocomputing, Neural Processing Letters, IEEE Signal Processing Letters, Neural Networks, IEEE Access, Expert Systems With Applications, Knowledge-Based Systems, AAAI, ECML PKDD, ICPR, ICME, ICIP, ICASSP, ICMR, EUSIPCO

Invited Talks

- 2023 **AI in Energy**, FemPower Summer School, Thessaloniki, Greece
2023 **Deep Learning for Electric Load Demand Forecasting**, Open Workshop “AI in Energy”, Thessaloniki, Greece
2022 **Deep Learning Techniques in Digital Media**, Hellenic Conference on Artificial Intelligence (SETN), Corfu, Greece

Participation in Conferences

- 2024 **IEEE International Symposium on Multimedia (ISM)**, Participation and presentation of 1 paper, Virtual-Tokyo, Japan
2024 **European Conference on Signal Processing (EUSIPCO)**, Participation and presentation of 1 paper, Lyon, France
2024 **International Conference on Engineering Applications of Neural Networks (EANN)**, Participation and presentation of 1 paper, Virtual-Corfu, Greece
2023 **International Conference on Engineering Applications of Neural Networks (EANN)**, Participation and presentation of 1 paper, Virtual-León, Spain
2022 **Panhellenic Conference on Electronics and Telecommunications (PACET)**, Participation and presentation of 2 papers, Tripolis, Greece
2022 **Hellenic Conference on Artificial Intelligence (SETN)**, Participation and presentation of the PhD thesis, Corfu, Greece
2022 **International Conference on Information, Intelligence, Systems & Applications (IISA)**, Participation and presentation of 2 papers, Corfu, Greece
2022 **IEEE Image, Video, and Multidimensional Signal Processing Workshop (IVMSP)**, Participation, presentation of 1 paper and chairing a session, Nafplio, Greece
2022 **Summit on Gender Equality in Computing (GEC)**, Participation and presentation of 2 papers, Thessaloniki, Greece
2021 **International Conference on Multimedia and Expo (ICME)**, Participation and presentation of 1 paper, Virtual-Shenzhen, China

- 2021 **International Conference on Information, Intelligence, Systems & Applications (IISA)**, Participation and presentation of 1 paper, Virtual-Crete, Greece
- 2020 **International Conference on Pattern Recognition (ICPR)**, Participation and presentation of 1 paper, Virtual-Milano, Italy
- 2020 **IEEE International Workshop on MACHINE LEARNING FOR SIGNAL PROCESSING (MLSP)**, Participation and presentation of 1 paper, Virtual-Espoo, Finland
- 2018 **Hellenic Conference on Artificial Intelligence (SETN)**, Participation and presentation of 2 papers, Patras, Greece
- 2016 **Hellenic Conference on Artificial Intelligence (SETN)**, Participation, Thessaloniki, Greece

Publications

Peer-reviewed Journals

1. O. Stentoumi, P. Nousi, **M. Tzelepi** and A. Tefas, "AnIO: Anchored Input-Output Learning for Time-series Forecasting", *Neural Computing and Applications*, 2024
2. **M. Tzelepi**, N. Passalis and A. Tefas, "Probabilistic Online Self-Distillation," *Neurocomputing*, 2022
3. E. Kakaletsis, C. Symeonidis, **M. Tzelepi**, I. Mademlis, A. Tefas, N. Nikolaidis, I. Pitas, "Computer Vision for Autonomous UAV Flight Safety: An Overview and a Vision-based Safe Landing Pipeline Example," *ACM Computing Surveys*, 2021
4. **M. Tzelepi**, N. Passalis and A. Tefas, "Online Subclass Knowledge Distillation," *Expert Systems with Applications*, 2021
5. **M. Tzelepi** and A. Tefas, "Improving the performance of lightweight CNNs for binary classification using Quadratic Mutual Information regularization," *Pattern Recognition*, 2020
6. N. Passalis, **M. Tzelepi** and A. Tefas, "Probabilistic Knowledge Transfer for Lightweight Deep Representation Learning," *IEEE Transactions on Neural Networks and Learning Systems*, 2020
7. **M. Tzelepi** and A. Tefas, "Class-Specific Discriminant Regularization in real-time deep CNN models for binary classification problems," *Neural Processing Letters*, 2019
8. **M. Tzelepi** and A. Tefas, "Graph Embedded Convolutional Neural Networks in Human Crowd Detection for Drone Flight Safety," *IEEE Transactions on Emerging Topics in Computational Intelligence*, 2019
9. E. Daskalakis, **M. Tzelepi** and A. Tefas, "Learning deep spatiotemporal features for video captioning," *Pattern Recognition Letters*, 2018
10. **M. Tzelepi** and A. Tefas, "Deep convolutional image retrieval: A general framework," *Signal Processing: Image Communication*, 2018
11. N. Kondylidis, **M. Tzelepi** and A. Tefas, "Exploiting tf-idf in deep Convolutional Neural Networks for Content Based Image Retrieval," *Multimedia Tools and Applications*, 2018
12. **M. Tzelepi** and A. Tefas, "Deep Convolutional Learning for Content Based Image Retrieval," *Neurocomputing*, 2018

Peer-reviewed Conferences (with proceedings)

1. **M. Tzelepi**, N. Dimitriou, C. Gkogkos and D. Tzovaras, "Improving zero-shot industrial defect detection exploiting LMMs as inverse reasoners", *IEEE International Conference on Image Processing (ICIP)*, 2026
2. **M. Tzelepi**, N. Dimitriou, C. Gkogkos and D. Tzovaras, "LMM-assisted Subclass-Aware Zero-Shot Industrial Defect Detection", *European Signal Processing Conference (EUSIPCO)*, 2026

3. **M. Tzelepi**, P. Nousi, N. Dimitriou and D. Tzovaras, "LMMs as Zero-Shot Teachers for Downstream Remote Sensing Classification Tasks", *Pattern Recognition in Remote Sensing (PRRS) Workshop at International Conference on Pattern Recognition (ICPR)*, 2026
4. P. Nousi and **M. Tzelepi**, "Improving Electric Load Demand Forecasting With Hard Representation Regularization", *European Signal Processing Conference (EUSIPCO)*, 2025
5. **M. Tzelepi** and V. Mezaris, "Improving multimodal hateful meme detection exploiting LMM-generated knowledge", *Multimodal Learning and Applications (MULA) Workshop at IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025
6. **M. Tzelepi** and V. Mezaris, "LMM-Regularized CLIP Embeddings for Image Classification", *IEEE International Symposium on Multimedia (ISM)*, 2024
7. A. Vafeidis, **M. Tzelepi** and A. Tefas, "Anchored Input-Output Encoding for Improving the Performance of Temporal Convolutional Networks", *IEEE International Conference on Information, Intelligence, Systems and Applications (IISA)*, 2024
8. **M. Tzelepi** and V. Mezaris, "Disturbing Image Detection Using LMM-Elicited Emotion Embeddings", *Integrating Image Processing with Large-Scale Language/Vision Models for Advanced Visual Understanding (LVLM) Workshop at IEEE International Conference on Image Processing (ICIP)*, 2024
9. **M. Tzelepi** and V. Mezaris, "Online Anchor-based Training for Image Classification Tasks", *IEEE International Conference on Image Processing (ICIP)*, 2024
10. **M. Tzelepi** and V. Mezaris, "Prototype Anchoring for Image Classification Tasks", *European Signal Processing Conference (EUSIPCO)*, 2024
11. **M. Tzelepi** and V. Mezaris, "Exploiting LMM-based knowledge for image classification tasks", *International Conference on Engineering Applications of Neural Networks (EANN)*, 2024
12. D. Moralis, **M. Tzelepi** and A. Tefas, "Retrieval-based methodology for few-sample logo recognition", *IEEE International Workshop on Multimedia Signal Processing (MMSP)*, 2023
13. A. Andronikos, **M. Tzelepi** and A. Tefas, "Residual Error Learning for Electricity Demand Forecasting", *International Conference on Engineering Applications of Neural Networks (EANN)*, 2023
14. **M. Tzelepi**, P. Nousi and A. Tefas, "Improving Electric Load Demand Forecasting with Anchor-based Forecasting Method", *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023
15. P. Nousi, **M. Tzelepi** and A. Tefas, "Anchored Input-Output Learning for Electrical Load Demand Forecasting", *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2023
16. **M. Tzelepi**, A. Sapountzaki, N. Maragkos and A. Tefas, "Online Self-Distillation for Electric Load Demand Forecasting on Greek Energy Market", *PAnhellenic Conference on Electronics and Telecommunications (PACET)*, 2022
17. G. Emmanouilidis, **M. Tzelepi** and A. Tefas, "Short-Term Electric Load Demand Forecasting on Greek Energy Market using Deep Learning: A comparative study", *PAnhellenic Conference on Electronics and Telecommunications (PACET)*, 2022
18. N. Passalis, S. Pedrazzi, R. Babuska,..., **M. Tzelepi** and A. Tefas, "OpenDR: An Open Toolkit for Enabling High Performance, Low Footprint Deep Learning for Robotics", *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022
19. **M. Tzelepi**, C. Symeonidis, N. Nikolaidis and A. Tefas, "Multilayer Online Self-Acquired Knowledge Distillation", *International Conference on Pattern Recognition (ICPR)*, 2022
20. N. Passalis, **M. Tzelepi**, P. Charitidis, S. Doropoulos, S. Vologianidis and A. Tefas, "Deep Video Stream Information Analysis and Retrieval: Challenges and Opportunities", *International Conference on Multimedia Information Processing and Retrieval (MIPR)*, 2022
21. N. Maragkos, **M. Tzelepi**, N. Passalis, A. Adamakos and A. Tefas, "Electric load demand forecasting on Greek Energy Market using lightweight neural networks", *IEEE Image, Video, and Multidimensional*

Signal Processing Workshop (IVMSP), 2022

22. **M. Tzelepi** and A. Tefas, "Forecasting day-ahead electric load demand on Greek Energy Market", *IEEE International Conference on Information, Intelligence, Systems and Applications (IISA)*, 2022
23. **M. Tzelepi**, C. Symeonidis, N. Nikolaidis and A. Tefas, "Real-time synthetic-to-real human detection for robotics applications", *IEEE International Conference on Information, Intelligence, Systems and Applications (IISA)*, 2022
24. **M. Tzelepi**, N. Tragkas and A. Tefas, "Improving binary semantic scene segmentation for robotics applications", *International Conference on Engineering Applications of Neural Networks (EANN)*, 2022
25. **M. Tzelepi** and A. Tefas, "Semantic Scene Segmentation for Robotics Applications", *IEEE International Conference on Information, Intelligence, Systems and Applications (IISA)*, 2021
26. **M. Tzelepi** and A. Tefas, "Efficient training of lightweight neural networks using Online Self-Acquired Knowledge Distillation", *IEEE International Conference on Multimedia and Expo (ICME)*, 2021
27. **M. Tzelepi**, N. Passalis and A. Tefas, "Efficient Online Subclass Knowledge Distillation for Image Classification", *International Conference on Pattern Recognition (ICPR)*, 2020
28. N. Passalis, **M. Tzelepi** and A. Tefas, "Heterogeneous Knowledge Distillation using Information Flow Modeling", *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020
29. N. Passalis, **M. Tzelepi** and A. Tefas, "Multilayer Probabilistic Knowledge Transfer for Learning Image Representations", *IEEE International Symposium on Circuits & Systems (ISCAS)*, 2020
30. **M. Tzelepi** and A. Tefas, "Quadratic Mutual Information Regularization in Real-Time Deep CNN Models", *IEEE International Workshop on Machine Learning for Signal Processing (MLSP)*, 2020
31. C. Nasioutzikis, **M. Tzelepi** and A. Tefas, "Deep Hashing Regularization Towards Hamming Space Retrieval", *Hellenic Conference on Artificial Intelligence (SETN)*, 2020
32. **M. Tzelepi** and A. Tefas, "Improving the Performance of Lightweight CNN Models Using Minimum Enclosing Ball Regularization", *European Signal Processing Conference, (EUSIPCO)*, 2019
33. **M. Tzelepi** and A. Tefas, "Discriminant analysis regularization in lightweight deep CNN models.", *IEEE International Conference on Image Processing (ICIP)*, 2019
34. E. Kakaletsis, **M. Tzelepi**, P. I. Kaplanoglou, C. Symeonidis, N. Nikolaidis, A. Tefas and I. Pitas, "Semantic Map Annotation through UAV Video Analysis using Deep Learning Models in ROS," *International Conference on MultiMedia Modeling (MMM)*, 2019
35. **M. Tzelepi** and A. Tefas, "Fully unsupervised optimization of CNN features towards Content Based Image Retrieval," *IEEE Image, Video, and Multidimensional Signal Processing (IVMSP) Workshop*, 2018
36. **M. Tzelepi** and A. Tefas, "Fully Unsupervised Convolutional Learning for Fast Image Retrieval," *Hellenic Conference on Artificial Intelligence (SETN)*, 2018 - **Among 5 Best Student Papers**
37. **M. Tzelepi** and A. Tefas, "Human Crowd Detection for Drone Flight Safety Using Convolutional Neural Networks," *European Signal Processing Conference (EUSIPCO)*, 2017
38. **M. Tzelepi** and A. Tefas, "Relevance feedback in deep convolutional neural networks for content based image retrieval," *Hellenic Conference on Artificial Intelligence (SETN)*, 2016
39. **M. Tzelepi** and A. Tefas, "Exploiting supervised learning for finetuning deep CNNs in content based image retrieval," *International Conference on Pattern Recognition (ICPR)*, 2016

Peer-reviewed Conferences (without proceedings)

1. **M. Tzelepi** and A. Tefas, "Electric Load Demand Forecasting on Greek Energy Market", *4th Summit on Gender Equality in Computing (GEC)*, 2022
2. P. Nousi, **M. Tzelepi**, N. Passalis and A. Tefas, "Lightweight Deep Learning Methods for Robotics Applications Using the OpenDR Toolkit", *4th Summit on Gender Equality in Computing (GEC)*,

2022

3. **M. Tzelepi**, C. Symeonidis, N. Nikolaidis and A. Tefas, "Multilayer Online Self-Acquired Knowledge Distillation", *4th Summit on Gender Equality in Computing (GEC)*, 2022

Book Chapters

1. P. Nousi, **M. Tzelepi**, N. Passalis and A. Tefas, "Lightweight Deep Learning", *Deep Learning for Robot Perception and Cognition, Edited by A. Iosifidis and A. Tefas*, 2022
2. N. Passalis, **M. Tzelepi** and A. Tefas, "Knowledge Distillation", *Deep Learning for Robot Perception and Cognition, Edited by A. Iosifidis and A. Tefas*, 2022
3. **M. Tzelepi**, P. Nousi, N. Passalis and A. Tefas, "Representation Learning and Retrieval", *Deep Learning for Robot Perception and Cognition, Edited by A. Iosifidis and A. Tefas*, 2022